简历（杨杰）

个人基本信息

姓名:	杨杰
性别:	男
出生年月:	1987 年 12 月 17 日
现工作单位:	西湖大学
个人学术主页：	**https://yangjie.ac.cn**
Google 学术主页：	**https://goo.gl/fqT63t**
电话:	+86-13157577935
E-mail:	**yangjie@westlake.edu.cn**

科研工作经历

2025 年 4 月-至今：脑机接口智能芯片系统浙江省工程研究中心，副主任

2025 年 4 月-至今：西湖大学，先进神经芯片中心，研究员

2019 年 4 月-2025 年 4 月：西湖大学，先进神经芯片中心，副研究员

2015 年 9 月-2019 年 2 月：卡尔加里大学（加拿大），博士后

教育背景

2010 年 9 月-2015 年 7 月：中国科学院半导体研究所，微电子学与固体电子学专业，工学博士

导师：刘剑研究员，吴南健研究员

2006 年 9 月-2010 年 7 月：天津大学电子信息工程学院，电子科学与技术专业，工学学士

个人荣誉

2026 年，国际电路与系统会议（ICCS）青年科学家奖

2026 年，中国脑机接口华瑞奖“杰出青年奖”

2025 年，脑机接口青百荟青年专家

2025 年，浙江省高层次人才培养计划青年人才

2024 年，神经芯片中心优秀员工

2023 年，九三学社浙江省委员会参政议政先进个人

2023 年，九三学社杭州市西湖区优秀社员

2022 年，杭州市西湖明珠工程海外高层次人才（杭州市人才分类 C 类）

2021 年，西湖大学工学院优秀员工

研究成果荣誉

2026 年　**全球脑机接口 × 医保创新场景大赛 • 创新创业奖**；获奖主体：西湖灵犀；项目：《植入式汉语言脑机通信系统》。

2026 年　**全球脑机接口 × 医保创新场景大赛 • 前沿技术突破奖**；获奖主体：西湖大学医学院附属杭州市第一人民医院“西湖脑机”团队；项目：《基于自研专用解码芯片与 sEEG 颅内电极的 AI 侵入式汉语失语康复闭环系统》。

2026 年　**半导体学报 • 当期封面**；论文：《面向边缘智能的神经形态计算芯片与部署》；通讯作者。

2025 年　**National Conference on Machine Learning and Nature-Inspired Computing • Best Poster Awards**；通讯作者。

2025 年　**第二届中国脑机智能大会 • 亮点墙报**；通讯作者。

2025 年　**ISCAS • Best Paper Award for BioCAS Track**；论文：《Efficient Self-Adaptive Pseudo-Resistor with Rapid Settling and High Linearity for Neurorecording Front-End Circuits》；通讯作者。

2023 年　**AICAS • Best Paper Award**；论文：《Binary is All You Need: Ultra-Efficient Arrhythmia Detection with a Binary-Only Compressive System》。

2023 年　**TBioCAS • Best Neuromorphic Paper**；论文：《A 510μW 0.738-mm^2 6.2-pJ/SOP Online Learning Multi-Topology SNN Processor with Unified Computation Engine in 40-nm CMOS》；通讯作者。

2023 年　**TBioCAS • Best Neuromorphic Paper**；论文：《A 0.99-to-4.38 μJ/class Event-Driven Hybrid Neural Network Processor for Full-Spectrum Neural Signal Analyses》；通讯作者。

2022 年　**JBHI • 当期封面**；论文：《NeuroSEE: A Neuromorphic Energy-Efficient Processing Framework for Visual Prostheses》；通讯作者。

项目经历

科技创新 2030-“脑科学与类脑研究”重大项目，课题负责人

课题经费 685 万，2022 年 8 月至 2027 年 7 月

浙江省“尖兵”研发攻关计划，项目负责人

项目经费 750 万，2024 年 1 月至 2026 年 12 月

浙江省重点研发计划（2021-01 至 2024-06），项目骨干

浙江省领军型创新创业团队（2021-01 至 2023-12），项目骨干

西湖大学光电研究院院立项目（2023-05-2026-04），项目骨干

研究领域

目前研究领域为**脑机接口与类脑计算**，主要包括以下**三个分支**：

1、侵入式脑机接口芯片与系统

2、类脑计算芯片与系统

3、神经编码与解码

教学经历

一、西湖大学工作期间（共同授课，2019 年 4 月-至今）

- Digital Circuits, 本科生课程，英文授课，Fall, 2023
- Hardware Description Language and System Simulation，博士生课程，英文授课，2021-2023
- Integrated Circuits Design，博士生课程，英文授课，Fall, 2023

二、卡尔加里大学博士后工作期间（共同授课，2015 年 7 月-2019 年 4 月）

- Integrated Micro and Nanotechnology Sensory Systems，硕士生课程，英文授课，2016-2017

研究经历

一、西湖大学工作期间（2019 年 4 月-至今）

■ 侵入式脑机接口芯片与系统：低功耗、小面积、高通量神经记录芯片

- 采用 40nm 工艺设计并流片测试，**实现了通道面积和功耗分别为 0.0032mm^2，1.38μW 的神经信号记录芯片，核心指标较国际先进水平提高 50%**，全面满足超高通量全植入脑机接口的面积和功耗需求。
- 该成果于 2023 年发表在**生物医疗芯片设计顶级期刊——IEEE TBioCAS（IEEE Transactions on Biomedical Circuits and Systems，JCR 一区）**，系统整体性能 2024 年发表在**集成电路设计顶级会议——IEEE CICC 2024（2024 IEEE Custom Integrated Circuits Conference，美国丹佛）**。
- 该成果已实现产业转化，**获得“大走廊杯”2023 杭州科创精英赛总决赛金奖（2/300）、2024 年获西湖英才计划 B 类支持。**

■ 侵入式脑机接口芯片与系统：用于闭环神经调控的低功耗生理信号处理芯片

- 采用 55nm 工艺设计并流片测试，芯片工作频率 20MHz，**首次提出并在 μW 级别功耗条件下实现个性化、智能闭环神经调控，芯片在癫痫预测上准确率超过 90%**，可显著提高现有闭环神经调控仪器的调控精准度和运行时间。
- 相关成果分别于 2021 年、2023 年和 2024 年发表在**生物医疗芯片设计顶级期刊——IEEE TBioCAS（IEEE Transactions on Biomedical Circuits and Systems，JCR 一区）**。

■ 侵入式脑机接口芯片与系统：面向低功耗闭环神经调控的信号处理算法

- **采用网络剪枝、量化方法将神经网络算法压缩至 50 kB 以下，并利用软硬件协同设计实现算法在芯片上的低功耗部署（<1 mW）**，为解决智能闭环神经调控算法与植入式芯片算力、功耗之间的矛盾提供了行之有效的方法。

- 相关成果分别于 2021 年和 2023 年发表在**生物医疗领域高水平期刊 IEEE TBME（IEEE Transactions on Biomedical Engineering，JCR 一区）**和**生物医疗芯片设计顶级 SCI 期刊——IEEE TBioCAS（IEEE Transactions on Biomedical Circuits and Systems，JCR 一区）。**

■ 类脑计算芯片与系统：用于穿戴式和植入式设备的类脑信号处理芯片

- 采用 40nm 工艺设计并流片测试，工作频率范围 1MHz-50MHz，**首次提出的可重构计算引擎与拓扑感知数据重用策略**，降低芯片功耗至 510μW，提高推理能效至 6.2pJ/SOP，在国际上较早展示了类脑芯片在穿戴式以及植入式系统中的巨大应用潜力。
- 该成果发表在**生物医疗芯片设计顶级 SCI 期刊——IEEE TBioCAS（IEEE Transactions on Biomedical Circuits and Systems，JCR 一区）**，以及 Frontiers in Neuroscience、ISCAS、BioCAS 等知名期刊和会议。

■ 类脑芯片与系统方向：大规模集成的高能效类脑芯片与系统

- 采用 40-nm CMOS 工艺设计并流片测试，工作频率范围 1MHz-200MHz，**算力 65.2TOPS/W，能效 0.078pJ/SOP，较世界先进水平提高 77.7%**，在图像以及动作识别任务上实现了**最高的准确率以及最低的能耗**。
- 该成果 2024 年发表在**集成电路设计顶级会议——IEEE CICC 2024（2024 IEEE Custom Integrated Circuits Conference，美国丹佛）**。该成果扩展期刊版本同时发表在**集成电路设计顶级期刊——JSSC （IEEE Journal of Solid-State Circuits，JCR 一区）。**

■ 神经编解码方向：脑-汉语解码

- 从汉语发音音节的声母、声调和韵母三个要素出发，**国际上首次在神经解码领域实现覆盖所有 407 个汉语拼音音节的脑-汉语解码范式**，在自由解码场景中，字符错误率中位数平均仅为 29%。
- 该成果 2023 年以预印本形式率先发表，**2025 年正式发表在生物医学领域顶刊 Cyborg and Bionic Systems（2024 年影响因子 18.13），2025 年在人工智能领域顶级会议 ICLR（2025 International Conference on Learning Representations，新加坡）**

■ 神经编解码方向：视网膜假体编码

- 首次采用仿生视觉传感与类脑计算技术，成功模拟了视网膜信息处理机制，**实现了预测脉冲发放率与真实视网膜的高拟合度（泊松系数超过 0.9），预测能耗显著降低至 10 毫焦级别**，开创了视网膜假体设计的新范式。
- 相关研究成果 2022 年作为**封面文章**发表在**生物医学信息领域顶级期刊 IEEE JBHI （IEEE Journal of Biomedical and Health Informatics，JCR 一区）**。2024 年在**人工智能领域顶会 ICLR（2024 International Conference on Learning Representations，奥地利维也纳）**发表。

二、卡尔加里大学博士后工作期间（2015 年 9 月-2019 年 2 月）

■ 高动态范围图像处理算法

- 提出的多重曝光的多尺度融合算法、基于拉普拉斯金字塔的高动态融合算法等，超越传统算法获得了超过 0.90 以上的图像品质指数，可以详细的展示图像中的高动态细节。

- 该成果 2017 年发表在信号处理领域会议 IEEE AVSS（IEEE International Conference on Advanced Video and Signal Based Surveillance）和 IEEE CAMSAP（International Workshop on Computational Advances in Multi-Sensor Adaptive Processing）。

■ 高动态范围图像处理硬件系统

- 设计并集成了高动态图像传感器和专用的高动态信号处理电路，采用所提出的尾数-指数算法，实现了百万像素传感器超过 100 帧的高动态渲染，具备了满足工业应用的条件。
- 相关研究成果 2019 年发表在**集成电路设计领域知名期刊 IEEE TCAS-II（IEEE Transactions on Circuits and Systems-II，JCR 二区）。**

三、中国科学院半导体研究所攻读博士学位期间（2010 年 9 月-2015 年 7 月）

■ 类脑人工智能视觉芯片

- 采用 65nm 标准 CMOS 工艺制程流片并测试，集成深度神经网络，**首次在视觉芯片中高效地完成包括完成人脸识别、目标检测等多种计算机视觉和人工智能任务**，芯片峰值运算性能达到 413GOPS。
- 该成果 2016 年发表在**视觉处理领域顶级期刊 IEEE TCSVT（IEEE Transactions on Circuits and Systems for Video Technology，JCR 一区）。**

■ 高速人工智能视觉芯片

- 采用 180nm 工艺设计流片并测试，**实现了国内最早的 1000fps 高速视觉芯片**，利用片上自组织神经网络实现特征提取与分类，满足高速场景视觉处理任务需求。
- 该成果 2014 年发表**在集成电路设计领域顶级期刊 IEEE JSSC（IEEE Transactions on Circuits and Systems for Video Technology，JCR 一区）。**

学术成果 (* 为通信作者，# 为共同第一作者)

书籍章节:

1. **杨杰**，Mohammed Sawan，《侵入式脑机接口芯片设计》，**“十四五”普通高等教育规划教材《脑机接口技术及其应用》第 22 章**，科学出版社，出版中。
2. Mohammed Sawan[#], Jie Yang[#] 等，《Emerging Trends of Biomedical Circuits and Systems》，Foundations and Trends® in Integrated Circuits and Systems。

期刊论文:

2027

71. S. Zhao, Z. Chen, Z. Wang, C. Fu, H. Ding, **J. Yang**, M. Sawan, "Dual-path continual learning with cross-scale attention for sparse physiological signal analysis", ***Biomedical Signal Processing and Control, vol. 130, Art. no. 111555, February 2027***, 2027.（正式卷期尚未到期）

2026

70. Z. Liao, C. Zhang, C. Fang, Y. Zhang, Z. Shen, **J. Yang***, M. Sawan*, " 面向边缘智能的神经形态计算芯片与部署", ***Chinese Journal of Semiconductors***, 2026.
69. X. Liu, H. Ali, J. Chen, H. Wu, W. Zou, **J. Yang***, M. Sawan*, "RF Energy Harvesting Integrated Circuits for Low-Power Sensors: A Review", ***IEEE Sensors Journal***, 2026.
68. Z. Shen, Z. Liao, S. Shen, C. Fang, F. Tian, **J. Yang***, M. Sawan*, "SHARP: An Energy-Efficient Spike-Based 3-D Recognition Processor With Hierarchical Hybrid Sparsity Awareness on Point Cloud", ***IEEE Transactions on Circuits and Systems I: Regular Papers***, 2026.
67. H. Liu, S. Zhao, C. Fu, **J. Yang**, M. Sawan, "A plug-and-play hybrid pruning framework for spike-driven transformers via adaptive spiking statistical importance scoring", ***Neurocomputing***, 2026.
66. S. Zhao, Q. Wang, Z. Shu, X. Yang, M. Fang, **J. Yang**, M. Sawan, "A Bioinspired Spiking Attention Neuromorphic Accelerator for Low-Power Visual Prosthesis", ***IEEE Transactions on Instrumentation and Measurement***, 2026.
65. W. Zou, **J. Yang***, M. Sawan*, "A Duty-Cycled PLL and Fractal Antenna Co-Design Architecture for a Low-Power IR-UWB Transmitter in Neural Implants", ***Sensors***, 2026.
64. W. Zou, R. Eskandari, **J. Yang***, M. Sawan*, "A High-Energy-Efficiency, Tunable-Bandwidth, and OOK IR-UWB Transmitter for Implantable Brain–Computer Interfaces", ***Electronics***, 2026.
63. H. Wu, J. Chen, R. Eskandari, Y. Han, X. Liu, W. Zou, F. Tian, Q. Hou, **J. Yang***, M. Sawan*, "NeuroSEED: A Neuromorphic Scalable Event-Driven SoC with Direct Multiplexing for Spike Detection and Wireless Telemetry", ***IEEE Journal on Emerging and Selected Topics in Circuits and Systems***, 2026.
62. S. Zhao, Z. Wang, H. Sun, S. Wang, Z. Chen, C. Fu, **J. Yang**, M. Sawan, "HAQ-ViT: A hardware-aware post-training quantization for efficient vision transformer inference", ***Knowledge-Based Systems***, 2026.
61. C. Zhang, Y. Zhang, J. Yu, **J. Yang***, M. Sawan*, "Event-Fused Hybrid ANN-SNN Architecture for Low-Latency Object Detection in Automotive Vision", ***IEEE Robotics and Automation Letters***, 2026.
60. C. Wang, D. Wu, Y. Liao, W. Zheng, Z. Zeng, X. Gao, H. Wu, Z. Zhu, **J. Yang**, L. Zhong, W. Cheng, Y-H. Chen, M. Sawan, "NeuroCLIP: A Multimodal Contrastive Learning Method for rTMS-treated Methamphetamine Addiction Analysis", ***IEEE Journal of Biomedical and Health Informatics***, 2026.

2025

59. H. Wu, Z. Tan, X. Liu, J. Chen, W. Zou, Q. Hou, S. Liu, Y. Mao, X. Kuang, **J. Yang***, M. Sawan*, "Self-Adaptive Pseudo-Resistors Enabling Millisecond-Level Artifact Recovery and High-Linearity for Neural Recording Front-Ends", ***IEEE Transactions on Biomedical Circuits and Systems***, 2025.
58. S. Li, D. Wu, S. Zhao, **J. Yang**, M. Sawan, "Memory-Efficient Intrinsic Gating Adaptation for Enhanced On-Device Epilepsy Diagnosis", ***IEEE Journal of Biomedical and Health Informatics***, 2025.
57. Z. Zeng, Y-H. Chen, X. Gao, W. Zheng, H. Wu, Z. Zhu, **J. Yang**, C. Wang, L. Zhong, W. Cheng, M. Sawan, "Repetitive TMS-based Identification of Methamphetamine-Dependent Individuals Using EEG Spectra", ***IEEE Sensors Journal***, 2025.
56. C. Fang, Z. Shen, T. Li, S. Zhao, F. Tian, **J. Yang***, M. Sawan*, " A 28nm Spiking Vision Transformer Accelerator with Dual-Path Sparse Compute Core and EMA-free Self-Attention Engine for Embodied Intelligence", ***IEEE Transactions on Circuits and Systems for Artificial Intelligence, 2025.***
55. Z. Wang, Y. Liao, **J. Yang***, M. Sawan*, "AI-driven and IC-enabled advanced closed-loop neuromodulation for practical clinical applications", ***Neuroelectronics***, 2025.

54. S. Zhao, S. Wang, Z. Wang, C. Fang, F. Tian, J. Chen, C. Fu, **J. Yang**, M. Sawan, "BoostViT: Booth-Serial Skipping and Tunable Scaling for Vision Transformers", ***IEEE Transactions on Circuits and Systems I: Regular Papers, 2025.***

53. X. Liu, J. Chen, H. Wu, W. Zou, **J. Yang***, M. Sawan*, "An On-Chip Reconfigurable Front-end for Ultra-Low-Power RF Energy Harvesting", ***IEEE Transactions on Circuits and Systems II: Express Briefs, 2025.***

52. S. Zhao, D. Wu, F. Tian, **J. Yang**, M. Sawan*, "A Resource-Efficient Algorithm-Hardware Co-Design Towards Semi-Supervised Neurological Symptoms Prediction", ***IEEE Transactions on Instrumentation and Measurement, 2025.***

51. X. Shao, **J. Yang**, M. Sawan, W. Cui, "Bridging biology and robotics: advancing submersible technology from robotic to live-fish models", ***Academia Engineering, 2025.***

50. C. Feng, L. Cao, E. Zhang, T. Wang, X. Jiang, J. Chen, H. Wu, S. Lin, Q. Hou, J. Zhu*, **J. Yang***, M. Sawan*, Y. Zhang*, "Acoustic Inspired Brain-to-Sentence Decoder for Logosyllabic Language", ***Cyborg and Bionic Systems, 2025.*** **（生物医学工程顶级期刊，JCR 一区，影响影子 >18）**

49. X. Gao, Y-H. Chen, Z. Zeng, W. Zheng, C. Chai, H. Wu, Z. Zhu, **J. Yang**, L. Zhong, H. Shen, M. Sawan, "Specific endophenotypes in EEG microstates for methamphetamine use disorder", ***Frontiers in Psychiatry, 2025.***

48. S. Shang, Y. Xu, M. Jin, C. Ni, H. Jiang, X. Wang, X. Wang, Y. Cheng, M. Wang, S. Zhang, B. Ji, **J. Yang**, G. Wang, M. Sawan, "Biocompatible dipeptide coated on Pt/PEDOT: PSS modified silicon probes for tissues rejection alleviation", ***Sensors and Actuators B: Chemical, 2025.***

47. X. Shen, Z. Zhang, **J. Yang**, J. Liu, N. Wu, M. Sawan, L. Liu, "A 0.0012-mm2 0.66-pJ/bit BPSK demodulator incorporating a loop-filter-less PLL achieving the maximum data rate of fcarrier/2", ***Journal of Semiconductors, 2025.***

2024

46. C. Fang, Z. Shen, Z. Wang, C. Wang, S. Zhao, F. Tian, **J. Yang***, M. Sawan*, "An Energy-Efficient Unstructured Sparsity-Aware Deep SNN Accelerator With 3-D Computation Array", ***IEEE Journal of Solid-State Circuits, 2024.*** **（集成电路设计领域顶级期刊，JCR 一区）**

45. F. Tian, J. Chen, J. Zheng, H. Wu, J. He, X. Wang, C. Fang, Y. Yao, **J. Yang**, M. Sawan, C-Y. Tsui, K-T. T. Cheng, "BioPI: An Energy Efficient and Low-Latency Neuromorphic Pipelined System with Joint Design Optimizations of Sensor-Algorithm-Processor for Wearable Healthcare", ***IEEE Transactions on Circuits and Systems for Artificial Intelligence, 2024.***

44. D. Wu, S. Li, **J. Yang***, M. Sawan*, "Neuro-BERT: Rethinking Masked Autoencoding for Self-Supervised Neurological Pretraining", ***IEEE Journal of Biomedical and Health Informatics, 2024.*** **（生物医学信息领域顶级期刊，JCR 一区）**

43. S. Zhao, C. Wang, C. Fang, F. Tian, **J. Yang***, M. Sawan*, "HybMED: A Hybrid Neural Network Training Processor with Multi-Sparsity Exploitation for Internet of Medical Things," ***IEEE Transactions on Biomedical Circuits and Systems***, 2024. **（生物医疗芯片设计顶级期刊，JCR 一区）**

42. **J. Yang**, S. Zhao, J. Wang, S. Lin, Q. Hou, M. Sawan, "Precise and low-power closed-loop neuromodulation through algorithm-integrated circuit co-design," ***Frontiers in Neuroscience***, 2024.

41. Y. Xu, **J. Yang**, W. Ming, S. Wang, M. Sawan, "Shorter latency of real-time epileptic seizure detection via probabilistic prediction," ***Expert Systems with Applications***, 2024. **（人工智能领域知名期刊，JCR 一区）**

2023

40. F. Xia, H. Li, Y. Li, X. Liu, Y. Xu, C. Fang, Q. Hou, S. Lin, Z. Zhang, **J. Yang**[*], M. Sawan, "Minimally Invasive Hypoglossal Nerve Stimulator Enabled by ECG Sensor and WPT to Manage Obstructive Sleep Apnea," ***MDPI Sensors***, 2023.
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邀请报告

28. 2026 脑机接口前沿技术与产业转化大会：《语言神经假肢：从高通量芯片到汉语神经解码》（2026 年 10 月 17 日）
27. ICCS 2026 邀请报告：《Integrated Circuits for Brain-Computer Interface From Lab to Clinic》（2026 年 9 月 20 日）
26. 全国软物质与生物物质物理学术会议：《高通量侵入式脑机接口系统级芯片：挑战与机遇》（2026 年 8 月 25 日）
25. 脑机接口开发者大会：《脑机接口芯片—脑信号读出与解码》（2026 年 8 月 15 日）
24. 6th International Workshop on Translation of Neurotechnologies：《Surviving the Cycle: Practices and Lessons Learned from Developing Speech Neuroprostheses》（2026 年 4 月 18 日）

23. 华人芯片设计技术研讨会 2026：《A 0.00022mm^2/electrode 1024 channel Sparity-Aware Neural Interface with CIM-Based Predictive Foucused Sampling for Hotspot Spike tracking》（2026 年 3 月 25 日）
22. 北方工业大学集成电路学院名师分享会：从芯片设计到语言解码看脑机接口的技术演进（2026 年 1 月 22 日）
21. 兰州大学物理学院系列报告：从芯片设计到语言解码看脑机接口的技术演进（2025 年 12 月 27 日）
20. 中国医疗器械行业协会产教融合发展大会:《语言神经假肢：从高集成芯片到汉语神经解码》（2025 年 11 月 29 日）
19. 脑机接口社区:《语言神经假肢：从高通量芯片到汉语神经解码》（2025 年 11 月 23 日）
18. 第四届南渡江论坛：报告主题:《脑机接口底层技术：从高集成芯片到汉语神经解码》（2025 年 11 月 11 日）
17. CCF DAC：报告主题:《类脑计算芯片、算法的协同设计及应用探索》（2025 年 11 月 2 日）
16. APCCAS：报告主题:《A Pipeline-Driven FPGA Accelerator ... & A Reconfigurable 8-Channel Dual-Mode Neural Stimulator ...》（2025 年 10 月 14 日）
15. 中国科学院半导体研究所物理沙龙：“脑机接口与类脑计算研究进展”（2025 年 3 月，中国北京）
14. 华人芯片设计技术研讨会 2025：“An Energy-Efficient Unstructured Sparsity-Aware Deep SNN Accelerator with 3D Compute Array”（2025 年 3 月，中国深圳）
13. 同济大学《微电子学堂》学术讲座：报告主题:《脑机接口——大脑信号读出与解码》（2024 年 11 月 21 日）
12. 华东师范大学 Neuroscience Guest Lecture Series：报告主题：《脑机接口—大脑信号读出与解码》（2024 年 10 月 18 日）
11. 2024 17th International Conference on Solid-State and Integrated Circuit Technology(ICSICT): “侵入式脑机接口的系统级芯片：挑战与机遇”（2024 年 10 月，中国珠海）
10. 昆山杜克大学神经科学专题研讨会：报告主题:《High-throughput brain-computer interface chip design and language decoding application》（2024 年 8 月 3 日）
9. 中国计算机学会芯片大会 (CCF Chip 2024): “Integrated Circuits for High Throughput Neural Recording”（2024 年 7 月，中国上海）
8. 九三学社中央第四十次科学座谈会：“学科交叉与产业投资双驱动——加速脑机接口技术由点到链的发展”（2024 年 5 月，中国天津）
7. 2024 片上脑机接口国际会议：“全植入脑机接口芯片与脑-汉语神经解码”（2024 年 4 月，中国天津）
6. 之江同心 • 经济圆桌会议：“脑机接口技术与应用”（2023 年 11 月，中国杭州）
5. 脑机接口与神经工程前沿交叉研讨会：“脑机接口芯片与汉语解码研究进展”（2023 年 11 月，中国杭州）

4. 脑机智能·未来科技创业沙龙：“脑机接口技术与应用”（2023 年 8 月，中国杭州）
3. 2022 20th IEEE International NEWCAS Conference: “Towards Smart and Closed-Loop Neuromodulation”(2022 年 6 月，加拿大魁北克，在线报告)
2. 浙江大学求是高等研究院学术讲座:“Intelligent Integrated Circuits for Biomedical Applications”（2021 年 5 月，中国杭州）
1. 中国科学院半导体研究所物理沙龙：“高动态图像处理”（2018 年 10 月，中国北京）

学术兼职

1、 学术期刊审稿人：IEEE Journal of Solid-State Circuits, IEEE Transaction on Industrial Electronics, IEEE Transactions on Biomedical Circuits and Systems, IEEE Transactions on Circuit and System for Video Technology, IEEE Transactions on Instrumentation & Measurement, IEEE Transactions on Neural Systems and Rehabilitation Engineering, IEEE Transactions on Biomedical Engineering, IEEE Transactions on Circuits and Systems I, IEEE Transactions on Circuits and Systems II, IEEE Transactions on Pattern Analysis and Machine Intelligence

2、 国际会议 2023 IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS 2023), 地方分会主席

3、 IEEE 高级会员